

Task 3: Pharmacovigilance & Adverse Drug Reaction (ADR) Analysis of Diclofenac

Introduction

Pharmacovigilance is the science of detecting, assessing, understanding, and preventing adverse effects or any other medicine-related problems. It plays an important role in ensuring that medicines are used safely and effectively. Adverse Drug Reactions (ADRs) are unwanted and harmful effects that occur when a medicine is taken at normal therapeutic doses. Monitoring and reporting ADRs help improve patient safety and support rational use of medicines.

Diclofenac is a non-steroidal anti-inflammatory drug (NSAID) commonly prescribed for pain, inflammation, fever, rheumatoid arthritis, osteoarthritis, ankylosing spondylitis, postoperative pain, and musculoskeletal disorders. It is one of the most widely used NSAIDs because of its strong anti-inflammatory and analgesic properties. However, long-term or inappropriate use can lead to serious adverse effects, making pharmacovigilance essential.

Drug Profile

- **Generic Name:** Diclofenac
- **Drug Class:** Non-Steroidal Anti-Inflammatory Drug (NSAID)
- **Dosage Forms:** Tablets, Capsules, Injection, Gel, Topical Patch, Suppository
- **Route of Administration:** Oral, Topical, Intramuscular, Intravenous, Rectal

Indications

Diclofenac is prescribed for:

- Rheumatoid arthritis
- Osteoarthritis
- Ankylosing spondylitis
- Acute musculoskeletal pain

- Postoperative pain
- Dental pain
- Dysmenorrhea (menstrual pain)
- Sports injuries

Mechanism of Action

Diclofenac works by inhibiting the Cyclooxygenase enzymes (COX-1 and COX-2), which are responsible for the production of prostaglandins. Prostaglandins are chemical mediators that cause pain, inflammation, and fever.

By reducing prostaglandin synthesis, diclofenac:

- Relieves pain
- Reduces inflammation
- Lowers fever
- Improves mobility in arthritis patients

Although effective, inhibition of COX-1 also decreases protective prostaglandins in the stomach and kidneys, increasing the risk of gastrointestinal and renal adverse effects.

Adverse Drug Reactions (ADRs)

Common ADRs

- Nausea
- Vomiting
- Gastric irritation
- Heartburn
- Abdominal pain
- Diarrhea
- Headache
- Dizziness
- Skin rash

- Edema (swelling)

Serious ADRs

- Gastrointestinal bleeding
 - Peptic ulcer
 - Acute kidney injury
 - Liver toxicity
 - Elevated liver enzymes
 - Myocardial infarction
 - Stroke
 - Severe allergic reactions (Anaphylaxis)
 - Stevens–Johnson Syndrome (rare)
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Risk Factors

The risk of ADRs increases in:

- Elderly patients
- Patients with kidney disease
- Patients with liver disease
- Patients with gastric ulcers
- Individuals with cardiovascular disease
- Patients receiving long-term NSAID therapy
- Patients taking anticoagulants or corticosteroids

Role of Pharmacovigilance

Pharmacovigilance helps in:

- Early detection of unknown ADRs
- Monitoring the safety of marketed medicines
- Improving patient care

- Reducing medication-related risks
- Supporting safe prescribing practices
- Identifying rare adverse reactions through continuous monitoring

Healthcare professionals should report suspected ADRs to national pharmacovigilance programs. Continuous reporting helps regulatory authorities evaluate drug safety and update treatment guidelines when necessary.

Prevention and Risk Reduction

The following measures help reduce ADRs associated with diclofenac:

- Use the lowest effective dose.
- Prescribe for the shortest possible duration.
- Monitor liver and kidney function during prolonged therapy.
- Use gastroprotective drugs (e.g., proton pump inhibitors) in high-risk patients.
- Avoid use in patients with active peptic ulcer disease or severe heart failure.
- Counsel patients to report symptoms such as black stools, severe abdominal pain, chest pain, breathing difficulty, or skin rashes immediately.

References

1. World Health Organization (WHO). *Safety Monitoring of Medicinal Products: Guidelines for Setting Up and Running a Pharmacovigilance Centre*.
2. Goodman & Gilman's *The Pharmacological Basis of Therapeutics*.
3. Katzung BG. *Basic & Clinical Pharmacology*.
4. Indian Pharmacopoeia.

5. Drug Bank Online – Diclofenac Monograph.